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MINDSPACE

LIBRARY FINE MANAGEMENT SYSTEM

ABSTRACT

MindSpace Library Fine Management System is a Java application developed to manage common library activities. It maintains student and book information, supports book issue and return, calculates delayed days, applies category-based fine calculation, and displays fine records. The project contains both a console implementation using Scanner and a Swing-based graphical interface.

The application stores its data locally in text files and uses Java classes to represent students, books, and library transactions. The project demonstrates object-oriented programming, conditional statements, switch-case, loops, methods, collections, date handling, file handling, Swing components, and event handling.

1. INTRODUCTION

A library needs to maintain accurate information about books, students, issued books, returned books, and fines. Manual record keeping can make it difficult to track availability and calculate overdue charges.

MindSpace provides a small-scale library management solution. The system keeps book availability updated, stores student and transaction information, calculates overdue days using dates, and generates fine information when a book is returned late.

2. PROBLEM STATEMENT

Develop a Library Fine Management System in Java that accepts student and book information, supports book issue and return operations, calculates delayed days, determines fines based on delayed days and book category, maintains multiple library records, and provides a fine receipt.

3. OBJECTIVES

- Maintain student and book information.
- Display and search the library book collection.
- Issue books to registered or newly entered students.
- Return issued books and update their availability.
- Calculate actual and delayed days using dates.
- Calculate fines using delayed days and book category.
- Maintain multiple library transaction records.
- Store and load data using local text files.
- Provide a graphical interface using Java Swing.
- Demonstrate fundamental Java programming concepts.

4. SCOPE OF THE SYSTEM

- Book collection and availability management.
- Student information management.
- Book issue and return processing.
- Overdue day calculation.
- Category-based fine calculation.
- Fine receipt and fine-record viewing.
- Local file-based persistence.
- Console and graphical user interface operation.

5. TECHNOLOGIES USED

Technology	Use in Project
Java	Core application development
Java Swing	Desktop graphical user interface
Scanner	Console input in Main.java
LocalDate / ChronoUnit	Issue, due, return and delayed-day calculation
ArrayList	Storage of students, books and library records
File I/O	Reading and writing local text files
Git / GitHub	Source-code version control

6. SYSTEM REQUIREMENTS

Hardware Requirements

- Computer or laptop
- Keyboard and display
- Minimum 4 GB RAM recommended

Software Requirements

- Java Development Kit (JDK)
- Java-compatible IDE or terminal
- Operating system capable of running Java Swing

7. SYSTEM DESIGN / ARCHITECTURE

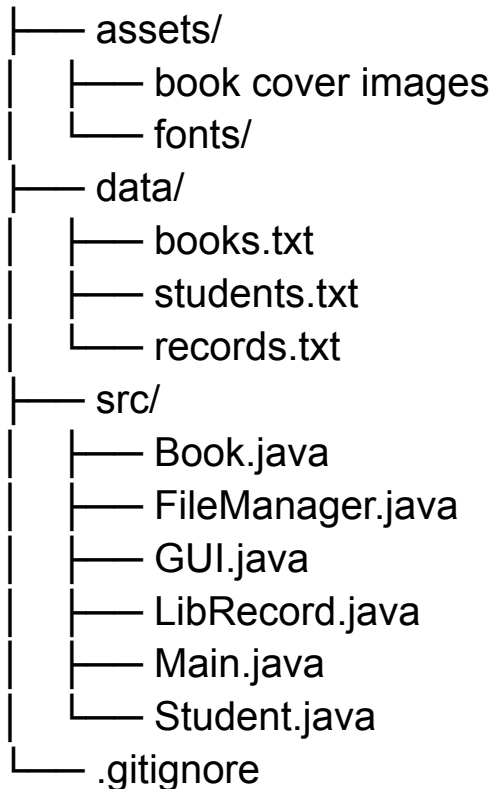
The project is organized into model classes, transaction logic, file-management logic, a console entry point, and a graphical interface.

Component	Responsibility
Student.java	Stores student ID, name, course and contact.
Book.java	Stores book information and manages total/available copies.
LibRecord.java	Stores issue/return transaction information and calculates fines.
FileManager.java	Creates, loads and saves books, students and library records.

Main.java	Provides the console-based library workflow using Scanner.
GUI.java	Provides the Swing dashboard and management screens.

8. PROJECT FILE STRUCTURE

LibraryFineManagementSystem/



9. FUNCTIONAL MODULES

9.1 Dashboard / Overview

Provides a central view of library information and navigation to the major modules.

9.2 Book Collection

Displays the book collection with cover images, search and category filtering.

9.3 Students

Displays student records and supports searching student information.

9.4 Issue Book

Allows a student and book to be selected, checks availability, and creates an active library record.

9.5 Return Book

Processes the return, calculates actual and delayed days, updates availability and calculates the applicable fine.

9.6 Fine Records

Displays recorded fine information and provides access to fine receipts.

9.7 Console Module

Main.java provides the Scanner-based console workflow required for the academic implementation.

10. FINE CALCULATION LOGIC

The implemented fine calculation first checks whether the delayed days are zero or less. If a delay exists, a switch statement selects a category multiplier and an if-else structure selects the delayed-day slab.

Delayed Days	Base Fine Rule
0 or less	₹0
1–7 days	Delayed days × ₹5
8–14 days	Delayed days × ₹10
15 or more	Delayed days × ₹20

The current category multipliers in LibRecord.java are Academic = 1.0, Reference = 1.5, General = 0.8, and Fiction = 1.0. These values are part of the current implementation and are used to demonstrate category-based fine calculation.

11. JAVA CONCEPTS USED

- Classes and Objects
- Encapsulation using private fields and getter methods
- Constructors
- Methods and method overloading
- if-else conditional statements
- switch-case
- Loops
- Arithmetic, comparison and logical operators
- ArrayList collections
- LocalDate and ChronoUnit
- File input/output
- Java Swing components
- ActionListener and event handling
- Basic GUI layout and custom drawing

12. DATA MANAGEMENT

The project uses local text files rather than a database. FileManager.java creates or prepares the required files and provides methods to load and save application data.

File	Stored Information
books.txt	Book ID, title, author, category, total copies and available copies
students.txt	Student ID, name, course and contact
records.txt	Student/book transaction, duration, fine, return status and date information

13. TESTING

The following test cases cover the major implemented operations.

Test Case	Action	Expected Result
TC01	Start application	Application loads the stored library data.
TC02	View book catalog	Book records and availability are displayed.
TC03	Search a book	Matching book records are displayed.
TC04	Filter book category	Books belonging to the selected category are displayed.
TC05	Issue available book	Book is issued and its availability is updated.
TC06	Issue unavailable book	Issue operation is prevented.
TC07	Return book on time	Book is returned with no overdue fine.
TC08	Return book late	Delayed days and category-based fine are calculated.
TC09	View fine receipt	Receipt displays transaction and fine information.
TC10	Restart application	Saved records can be loaded again from the text files.

14. USER INTERFACE SCREENSHOTS

Figure 1: Dashboard / Overview

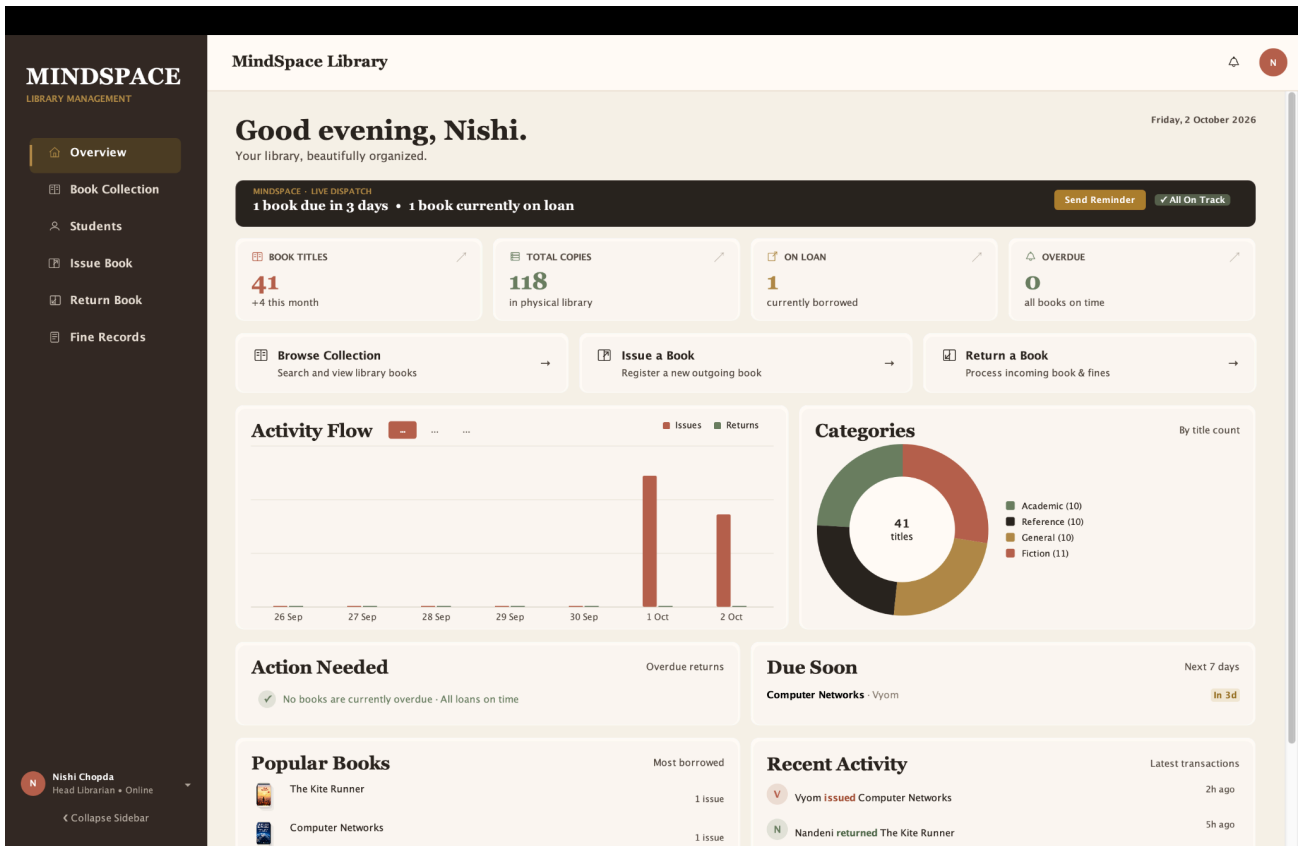


Figure 2: Book Collection

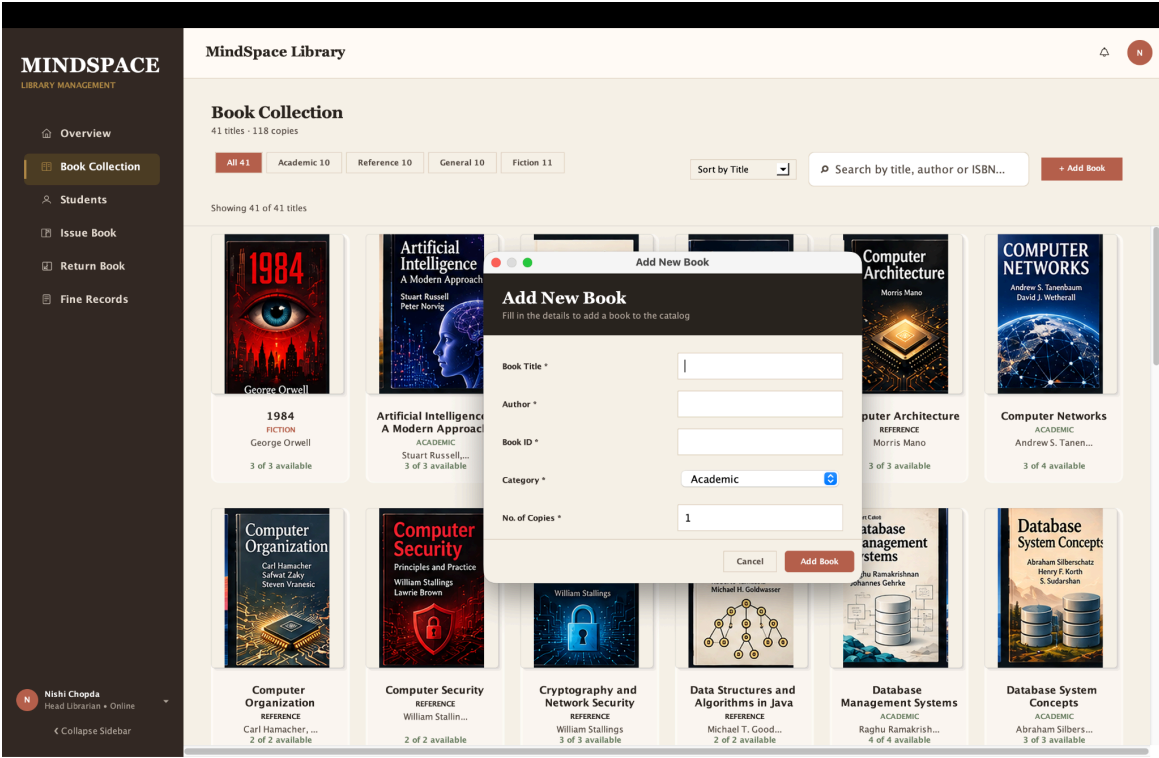
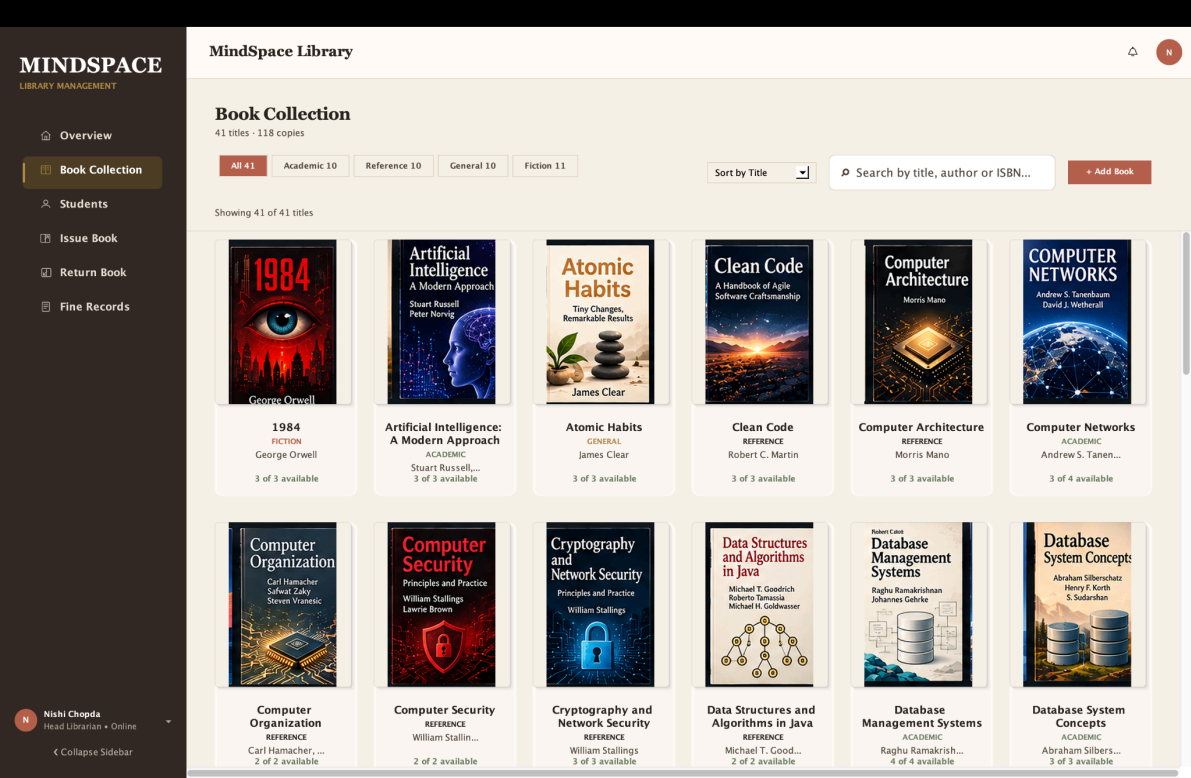


Figure 3: Students

MINDSPACE

LIBRARY MANAGEMENT

Overview

Book Collection

Students

Issue Book

Return Book

Fine Records

N Nishi Chopda

Head Librarian • Online

◀ Collapse Sidebar

MindSpace Library

Students

10 registered students

🔍 Search by name, ID or phone...

+ Add Student

All 10

With issued books 1

With fines 0

Overdue 0

Showing 10 of 10 students

STUDENT	COURSE	PHONE	ISSUED	FINE DUE	STATUS
N Nishi Chopda ST101	B.Tech CSE	9876543210	—	No dues	Active
N Nandeni Tiwari ST102	B.Tech CSE	9876543211	—	No dues	Active
A Akshara Tant... ST103	B.Tech CSE	9876543212	—	No dues	Active
Y Yashika Saraf ST104	B.Tech CSE	9876543213	—	No dues	Active
P Paarshvi Vijay ST105	B.Tech CSE	9876543214	—	No dues	Active
V Vyom Shah ST106	B.Tech CSE	9876543215	1 book	No dues	Active
K Kushal Nakt... ST107	B.Tech CSE	9876543216	—	No dues	Active
P Prathamesh ... ST108	B.Tech CSE	9876543217	—	No dues	Active
D Daksh Shriva... ST109	B.Tech CSE	9876543218	—	No dues	Active

MINDSPACE

LIBRARY MANAGEMENT

Overview

Book Collection

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Fine Records

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Head Librarian • Online

◀ Collapse Sidebar

MindSpace Library

Students

10 registered students

🔍 Search by name, ID or phone...

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All 10

With issued books 1

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Showing 10 of 10 students

STUDENT	COURSE	PHONE	ISSUED	FINE DUE	STATUS
N Nishi Chopda ST101	B.Tech CSE			No dues	Active
N Nandeni Tiwari ST102	B.Tech CSE			No dues	Active
A Akshara Tant... ST103	B.Tech CSE			No dues	Active
Y Yashika Saraf ST104	B.Tech CSE			No dues	Active
P Paarshvi Vijay ST105	B.Tech CSE			No dues	Active
V Vyom Shah ST106	B.Tech CSE			No dues	Active
K Kushal Nakt... ST107	B.Tech CSE	9876543216	—	No dues	Active
P Prathamesh ... ST108	B.Tech CSE	9876543217	—	No dues	Active
D Daksh Shriva... ST109	B.Tech CSE	9876543218	—	No dues	Active

Add Student

Name

Student ID

Course

Year

Phone

Cancel

Save

Figure 4: Issue Book

MINDSPACE
LIBRARY MANAGEMENT

Overview

Book Collection

Students

Issue Book

Return Book

Fine Records

N Nishi Chopda
Head Librarian • Online
Collapse Sidebar

MindSpace Library

Issue a Book
Create a new library issue record.

Issue details

Select Student
ST101 - Nishi Chopda • B.Tech CSE

Select Book
To Kill a Mockingbird — Harper Lee (3 of 3 available)

Issue Duration (Days)
7 days 14 days 21 days 30 days Custom 5

Issue Book

Live Issue Preview



To Kill a Mockingbird
by Harper Lee (Fiction)
Copies: 3 of 3 -> 2 of 3 after issue

N Nishi Chopda
ST101 - B.Tech CSE

Issue Date: 2 Oct 2026 (Today)
Due Date: 16 Oct 2026 (14 days loan)
Fine rule: 1-7 days: ₹5/day, 8-14: ₹10/day, 15+ ₹20/day

Recent Issues

The Book Thief → Nishi Chopda
ID: ST101 - B.Tech CSE

Issued: 2 Oct
Due: 7 Oct

Computer Networks → Vyom Shah
ID: ST106 - B.Tech CSE

Issued: 2 Oct
Due: 5 Oct

Figure 5: Return Book

MINDSPACE
LIBRARY MANAGEMENT

Overview

Book Collection

Students

Issue Book

Return Book

Fine Records

N Nishi Chopda
Head Librarian • Online
Collapse Sidebar

MindSpace Library

Return a Book
Process returned books and calculate fines.

CURRENTLY ON LOAN
2
books awaiting return

AVAILABLE BOOKS
116
copies in the library

RECORDED FINES
37
total fines collected (₹)

Return details

Select Active Loan
ST101 - Nishi Chopda — The Book Thief (due 7 Oct)

Return Date
2 October 2026 -1d +1d

DAYS KEPT 0
DAYS LATE 0

Return Book

Live Fine Preview

Returned on time, no fine



The Book Thief
Nishi Chopda (ST101)
Fiction

Issue Date: 2 Oct 2026
Due Date: 7 Oct 2026
Return Date: 2 Oct 2026
Days Late: 0 days
Slab Applied: On Time (No slab)
Rate per Day: ₹0 / day

TOTAL FINE
₹0

Recent Returns

The Kite Runner ← Nandani Tiwari
Returned on 3 Oct 2026 • Student ID: ST102

₹0

Java: The Complete Reference ← Nishi Chopda
Returned on 16 Oct 2026 • Student ID: ST101

₹37

Figure 6: Fine Records

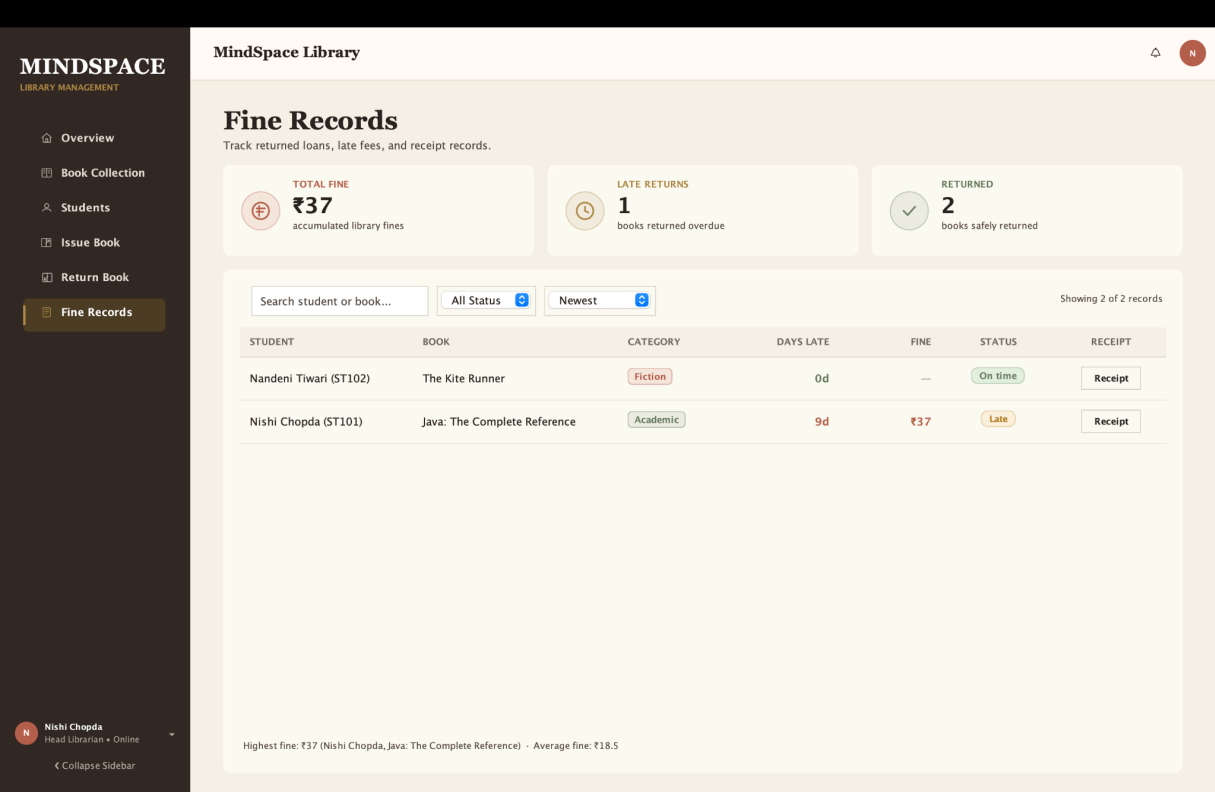
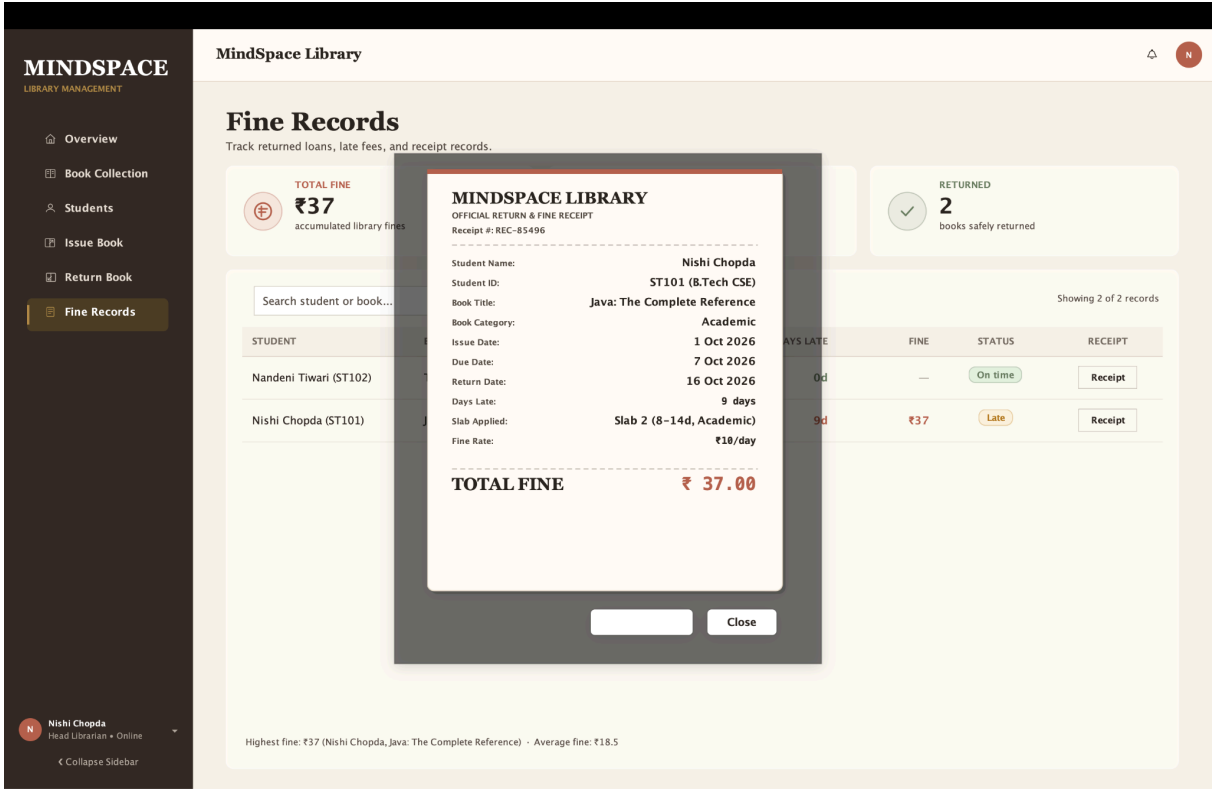


Figure 7: Fine Receipt



15. ADVANTAGES

- Simple library workflow.
- Automatic tracking of book availability.
- Automatic overdue-day calculation.
- Category-based fine calculation.
- Supports multiple students and library records.
- Local data persistence without requiring a database.
- Provides both console and graphical interfaces.

16. LIMITATIONS

- Data is stored in local text files instead of a database.
- The application is designed for a local desktop environment.
- Authentication and multi-user access control are not implemented.
- Fine rules are fixed in the Java implementation.

17. FUTURE SCOPE

- Database integration using MySQL or another database system.
- User login and role-based access.
- Automated overdue notifications.
- Online library access.
- Reports and analytics.
- Backup and restore functionality.

18. CONCLUSION

MindSpace Library Fine Management System provides a practical Java implementation of a small library workflow. The system covers book and student management, book issue and return, delayed day calculation, category-based fine calculation, fine records, and local file persistence. The project also demonstrates important Java concepts through both a console implementation and a Swing-based graphical interface.

19. REFERENCES

- Java programming concepts and standard Java library documentation.
- Course notes and faculty-provided study material.
- Project source code and local project data files.